Customer Risk & Claims Analysis for an Insurance Company

A report on the Fraud Analysis of Claims from customers of an Insurance Company

Project Overview & Background

- This project analyzes the claims of an insurance company that is facing:
  - Increasing claim costs
  - Suspected fraudulent claims
  - Low customer retention

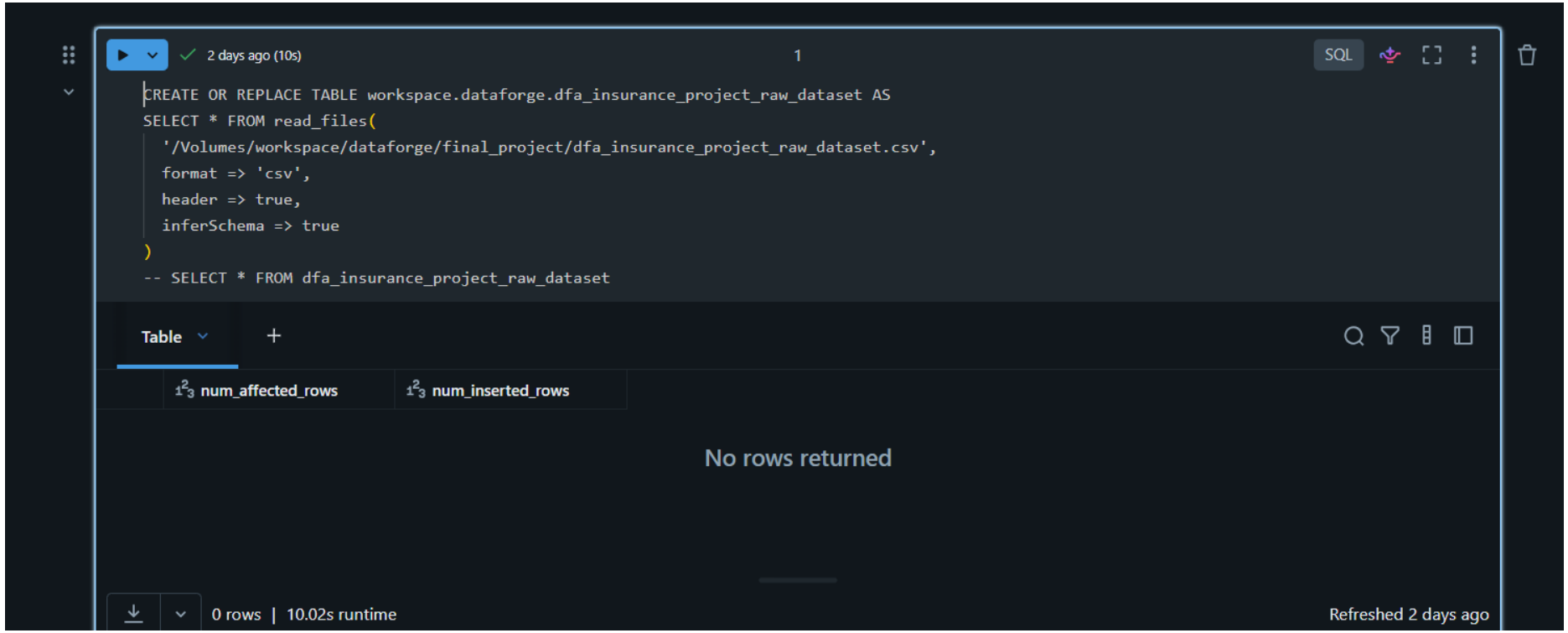
- The tools used:
  - SQL (Databricks)
  - Power BI (Dashboard)
  - Power Point (Presentation)

Dataset Structure

- Customer_ID – Unique customer
- Age – Customer age
- Gender – Male/Female
- Province – Province/City
- Monthly_Income – Customer income
- Join_Date – Date that customer joined policy
- Policy_ID – Policy identifier
- Policy_Type – Auto / Health / Life
- Premium Amount – Monthly premium
- Policy_Status – Active / Cancelled
- Claim_ID – Claim identifier
- Claim_Date – Date of claim
- Claim_Amount – Amount claimed
- Claim_Status – Approved / Rejected
- Fraud_Flag – Yes / No

Data Exploration

- The insurance dataset was uploaded onto Databricks and was processed using SQL in a notebook
- The dataset was used to create a table in a database using SQL



Data Exploration Questions

The Following questions were answered in the data exploration notebook:

- ❑Total number of customers
- ❑Total number of policies
- ❑Total number of claims
- ❑Total claim amount
- ❑Claims by policy type
- ❑Average claim amount per policy type
- ❑Number of fraud vs non-fraud claims
- ❑Top 5 highest claim amounts
- ❑Claims per location
- ❑Claims trend over time

Total number of customers

17 hours ago (1s) 3

```
SELECT Count(DISTINCT(Customer_ID)) AS Total_number_of_customers FROM workspace.dataforge.dfa_insurance_project_raw_dataset
```

See performance (1) Optimize

Table +

	Total_number_of_customers
1	518

Total number of customers is 518 customers

Total number of policies

17 hours ago (1s) 4

```
SELECT COUNT(DISTINCT(TRIM(Policy_ID))) AS Total_number_of_Policies FROM workspace.dataforge.dfa_insurance_project_raw_dataset
```

See performance (1) Optimize

Table +

	Total_number_of_Policies
1	520

Total number of policies is 520 policies

Total number of claims

17 hours ago (1s) 5

```
SELECT COUNT(Claim_ID) AS Total_number_of_Claims FROM workspace.dataforge.dfa_insurance_project_raw_dataset
```

See performance (1) Optimize

Table

	Total_number_of_Claims
1	547

Total number of claims is 547 claims

Total claim amount

17 hours ago (3s) 6

```
SELECT
SUM(CAST(TRIM(coalesce(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%-%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '-', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE)
END,
(SELECT mean(Trim(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE)
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS DOUBLE)) AS Total_Claim_Amount FROM workspace.dataforge.
dfa_insurance_project_raw_dataset
```

See performance (1) Optimize

Table +

	1.2 Total_Claim_Amount
1	12578657.020740738

Total Claim Amount is R12 578 657.02

Claims by policy type

Home policy has 94 claims

Auto policy has 128 claims

Health policy has 134 claims

Motor policy has 10 claims

Life policy has 105 claims

Funeral policy has 68 claims

Medical policy has 5 claims

Property policy has 3 claims

17 hours ago (1s) 7

```
SELECT DISTINCT(TRIM(UPPER(Policy_Type))) As Policy_Type, count(TRIM(UPPER(Claim_ID))) AS Total_number_of_Policies_Claims FROM workspace.
dataforge.dfa_insurance_project_raw_dataset GROUP BY TRIM(UPPER(Policy_Type))
```

See performance (1) Optimize

Table

	Policy_Type	Total_number_of_Policies_Claims
1	HOME	94
2	AUTO	128
3	HEALTH	134
4	MOTOR	10
5	LIFE	105
6	FUNERAL	68
7	MEDICAL	5
8	PROPERTY	3

8 rows | 0.67s runtime Refreshed 17 hours ago

Average claim amount per policy type

Funeral policies claimed R14198.58

Home policies claimed R25467.26

Medical policies claimed R17714.30

Motor policies claimed R14929.18

Health policies claimed R12234.46

Life policies claimed R45518.10

Auto policies claimed R19049.83

Property policies claimed R20201.64

17 hours ago (2s) 8: Cell 7

```sql
SELECT TRIM(UPPER(Policy_Type)) AS Policy_Type,
MEAN(CAST(TRIM(coalesce(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%-%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '-', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE)
END,
(SELECT mean(Trim(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE)
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS DOUBLE)) AS Average_claim_amount_per_policy_type

 FROM workspace.dataforge.dfa_insurance_project_raw_dataset GROUP BY TRIM(UPPER(Policy_Type))
```

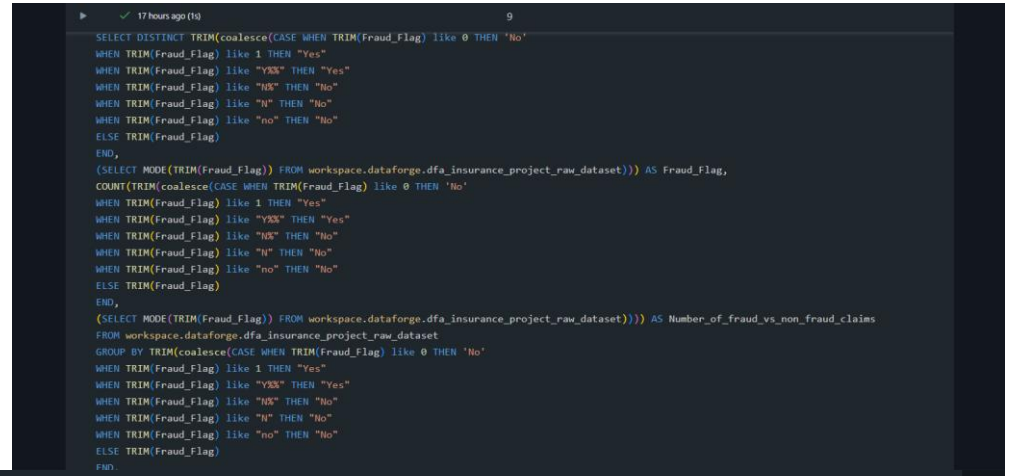
See performance (1) Optimize

Table

	Policy_Type	Average_claim_amount_per_policy_type
1	FUNERAL	14198.578531590416
2	HOME	25467.256000779722
3	MEDICAL	17714.302
4	MOTOR	14929.177
5	HEALTH	12234.462611940296

Number of fraud vs non-fraud claims

79 fraudulent claims
471 non-fraudulent claims



17 hours ago (1s) 9

```
SELECT DISTINCT TRIM(coalesce(CASE WHEN TRIM(Fraud_Flag) like 0 THEN 'No'
WHEN TRIM(Fraud_Flag) like 1 THEN "Yes"
WHEN TRIM(Fraud_Flag) like "Y%%" THEN "Yes"
WHEN TRIM(Fraud_Flag) like "N%" THEN "No"
WHEN TRIM(Fraud_Flag) like "N" THEN "No"
WHEN TRIM(Fraud_Flag) like "no" THEN "No"
ELSE TRIM(Fraud_Flag)
END,
(SELECT MODE(TRIM(Fraud_Flag)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS Fraud_Flag,
COUNT(TRIM(coalesce(CASE WHEN TRIM(Fraud_Flag) like 0 THEN 'No'
WHEN TRIM(Fraud_Flag) like 1 THEN "Yes"
WHEN TRIM(Fraud_Flag) like "Y%%" THEN "Yes"
WHEN TRIM(Fraud_Flag) like "N%" THEN "No"
WHEN TRIM(Fraud_Flag) like "N" THEN "No"
WHEN TRIM(Fraud_Flag) like "no" THEN "No"
ELSE TRIM(Fraud_Flag)
END,
(SELECT MODE(TRIM(Fraud_Flag)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS Number_of_fraud_vs_non_fraud_claims
FROM workspace.dataforge.dfa_insurance_project_raw_dataset
GROUP BY TRIM(coalesce(CASE WHEN TRIM(Fraud_Flag) like 0 THEN 'No'
WHEN TRIM(Fraud_Flag) like 1 THEN "Yes"
WHEN TRIM(Fraud_Flag) like "Y%%" THEN "Yes"
WHEN TRIM(Fraud_Flag) like "N%" THEN "No"
WHEN TRIM(Fraud_Flag) like "N" THEN "No"
WHEN TRIM(Fraud_Flag) like "no" THEN "No"
ELSE TRIM(Fraud_Flag)
END
```

17 hours ago (1s) 9

```
WHEN TRIM(Fraud_Flag) like 1 THEN "Yes"
WHEN TRIM(Fraud_Flag) like "Y%%" THEN "Yes"
WHEN TRIM(Fraud_Flag) like "N%" THEN "No"
WHEN TRIM(Fraud_Flag) like "N" THEN "No"
WHEN TRIM(Fraud_Flag) like "no" THEN "No"
ELSE TRIM(Fraud_Flag)
END,
(SELECT MODE(TRIM(Fraud_Flag)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset)))
```

See performance (1) Optimize

Table +

	Fraud_Flag	Number_of_fraud_vs_non_fraud_claims
1	Yes	79
2	No	471

Claims per location

Gauteng has 64 claims
Limpopo has 47 claims
Eastern Cape 75 claims
North West has 72 claims
Free State has 72 claims
Mpumalanga has 55 claims
Western Cape has 75 claims
KwaZulu-Natal has 90 claims

17 hours ago (2s) 11

```
CREATE OR REPLACE TEMPORARY VIEW Reference AS
SELECT TRIM(coalesce(CASE
WHEN Trim(UPPER(Province)) IN ("GP", "GAUTENG") THEN "Gauteng"
WHEN Trim(UPPER(Province)) IN ("EC", "EASTERN CAPE") THEN "Eastern Cape"
WHEN Trim(UPPER(Province)) IN ("FS", "FREE STATE") THEN "Free State"
WHEN Trim(UPPER(Province)) IN ("NW", "NORTH WEST") THEN "North West"
WHEN Trim(UPPER(Province)) IN ("WC", "WESTERN CAPE") THEN "Western Cape"
WHEN Trim(UPPER(Province)) IN ("MP", "MPUMALANGA") THEN "Mpumalanga"
WHEN Trim(UPPER(Province)) IN ("LP", "LIMPOPO") THEN "Limpopo"
WHEN Trim(UPPER(Province)) IN ("KZN", "KWAZULU-NATAL", "KWAZULU NATAL") THEN "KwaZulu-Natal"
END,
(SELECT mode(trim(CASE
WHEN Trim(UPPER(Province)) IN ("GP", "GAUTENG") THEN "Gauteng"
WHEN Trim(UPPER(Province)) IN ("EC", "EASTERN CAPE") THEN "Eastern Cape"
WHEN Trim(UPPER(Province)) IN ("FS", "FREE STATE") THEN "Free State"
WHEN Trim(UPPER(Province)) IN ("NW", "NORTH WEST") THEN "North West"
WHEN Trim(UPPER(Province)) IN ("WC", "WESTERN CAPE") THEN "Western Cape"
WHEN Trim(UPPER(Province)) IN ("MP", "MPUMALANGA") THEN "Mpumalanga"
WHEN Trim(UPPER(Province)) IN ("LP", "LIMPOPO") THEN "Limpopo"
WHEN Trim(UPPER(Province)) IN ("KZN", "KWAZULU-NATAL", "KWAZULU NATAL") THEN "KwaZulu-Natal"
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS Province
FROM workspace.dataforge.dfa_insurance_project_raw_dataset;

SELECT DISTINCT Province, COUNT(Province) AS Claims_per_location
FROM Reference GROUP BY Province
```

See performance (2)

17 hours ago (2s) 11

```
WHEN Trim(UPPER(Province)) IN ("LP", "LIMPOPO") THEN "Limpopo"
WHEN Trim(UPPER(Province)) IN ("KZN", "KWAZULU-NATAL", "KWAZULU NATAL") THEN "KwaZulu-Natal"
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS Province
FROM workspace.dataforge.dfa_insurance_project_raw_dataset;

SELECT DISTINCT Province, COUNT(Province) AS Claims_per_location
FROM Reference GROUP BY Province
```

See performance (2)

Show 1 additional result

Table

	Province	Claims_per_location
1	Gauteng	64
2	Limpopo	47
3	Eastern Cape	75
4	North West	72
5	Free State	72
6	Mpumalanga	55
7	Western Cape	75
8	KwaZulu-Natal	90

Top 5 highest claim amounts

- CUST1050 claimed R143857.60
- CUST1324 claimed R115579.76
- CUST1429 claimed R97527.12
- CUST1425 claimed R92918.98
- CUST1068 claimed R89909.74

11:22 AM (2s) 13

```sql
SELECT DISTINCT Customer_ID,
SUM(CAST(TRIM(coalesce(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%-%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '-', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE) END,
(SELECT mean(Trim(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%-%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '-', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE)
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS DOUBLE)) AS Claim_Amount
FROM workspace.dataforge.dfa_insurance_project_raw_dataset
GROUP BY Customer_ID
Order by CAST(TRIM(coalesce(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%-%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '-', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE) END,
(SELECT mean(Trim(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%-%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '-', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE)
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS DOUBLE) DESC
LIMIT 5
```

See performance (1) Optimize

Table

	Customer_ID	Claim_Amount
1	CUST1050	143857.6
2	CUST1324	115579.76
3	CUST1429	97527.12
4	CUST1425	92918.98
5	CUST1068	89909.74

5 rows | 2.01s runtime Refreshed 3 minutes ago

Claims trend over time

Claim over time has mainly single claims with a few multiple claims on some dates.

Considering that there are dates with multiple claims, we could say that the claims are increasing on an upward trend overtime.

10 hours ago (2s) 13

```sql
SELECT
CASE
WHEN substring(trim(Join_Date),3,1) = "/" THEN to_date(trim(Join_Date), 'dd/MM/yyyy')
WHEN substring(trim(Join_Date),5,1) = "-" THEN to_date(trim(Join_Date), 'yyyy-MM-dd')
WHEN substring(trim(Join_Date),5,1) = "/" THEN to_date(trim(Join_Date), 'yyyy/MM/dd')
WHEN trim(Join_Date) LIKE '__-___-____' THEN to_date(trim(Join_Date), 'dd-MMM-yyyy')
WHEN substring(trim(Join_Date),3,1) = "-" THEN to_date(trim(Join_Date), 'MM-dd-yyyy')
END AS Join_Date,
COUNT(CASE
WHEN substring(trim(Join_Date),3,1) = "/" THEN to_date(trim(Join_Date), 'dd/MM/yyyy')
WHEN substring(trim(Join_Date),5,1) = "-" THEN to_date(trim(Join_Date), 'yyyy-MM-dd')
WHEN substring(trim(Join_Date),5,1) = "/" THEN to_date(trim(Join_Date), 'yyyy/MM/dd')
WHEN trim(Join_Date) LIKE '__-___-____' THEN to_date(trim(Join_Date), 'dd-MMM-yyyy')
WHEN substring(trim(Join_Date),3,1) = "-" THEN to_date(trim(Join_Date), 'MM-dd-yyyy')
END) AS  Claims_trend_over_time
FROM workspace.dataforge.dfa_insurance_project_raw_dataset
GROUP BY CASE
WHEN substring(trim(Join_Date),3,1) = "/" THEN to_date(trim(Join_Date), 'dd/MM/yyyy')
WHEN substring(trim(Join_Date),5,1) = "-" THEN to_date(trim(Join_Date), 'yyyy-MM-dd')
WHEN substring(trim(Join_Date),5,1) = "/" THEN to_date(trim(Join_Date), 'yyyy/MM/dd')
WHEN trim(Join_Date) LIKE '__-___-____' THEN to_date(trim(Join_Date), 'dd-MMM-yyyy')
WHEN substring(trim(Join_Date),3,1) = "-" THEN to_date(trim(Join_Date), 'MM-dd-yyyy')
END
ORDER BY Join_Date ASC
```

See performance (1) Optimize

10 hours ago (2s) 13

ORDER BY Join_Date ASC

See performance (1) Optimize

Table

	Join_Date	Claims_trend_over_time
1	2020-02-29	1
2	2020-05-05	1
3	2020-05-18	1
4	2020-05-27	1
5	2020-05-30	1
6	2020-06-04	1
7	2020-06-14	1
8	2020-06-30	2
9	2020-07-08	1
10	2020-07-15	1
11	2020-07-30	1
12	2020-08-20	1
13	2020-08-22	1
14	2020-08-30	1
15	2020-09-12	1

Data Cleaning

- The dataset was initially very messy, presenting issue such as:
  - Missing values
  - Blank values
  - Text casing inconsistencies
  - Unnecessary spaces
  - Inconsistent date formatting
  - Inconsistent numeric formatting
  - Inconsistent data types
  - Redundant naming such as different terms with same meaning

SQL Data Cleaning

The dataset that was ingested on Databricks is cleaned in a separate SQL notebook and it creates a new cleaned dataset.

Customer_ID cleaning:

- Removed unnecessary spaces

Gender cleaning:

- Removed unnecessary spaces
- Replaced missing values
- Standardized the values

Age cleaning:

- Removed unnecessary spaces
- Changed data type to INT

24 hours ago (41s) 1 SQL

```sql
CREATE OR REPLACE TABLE workspace.dataforge.clean_insurance_data AS
SELECT
TRIM(Customer_ID) AS Customer_ID,

CAST(TRIM(Age) AS INT) AS Age,

TRIM(coalesce(CASE WHEN TRIM(Gender) like 'M%' THEN 'Male'
WHEN TRIM(Gender) like 'F%' THEN 'Female'
ELSE TRIM(Gender)
END,
(SELECT MODE(TRIM(Gender)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS Gender,

TRIM(coalesce(CASE
WHEN Trim(UPPER(Province)) IN ("GP", "GAUTENG") THEN "Gauteng"
WHEN Trim(UPPER(Province)) IN ("EC", "EASTERN CAPE") THEN "Eastern Cape"
WHEN Trim(UPPER(Province)) IN ("FS", "FREE STATE") THEN "Free State"
WHEN Trim(UPPER(Province)) IN ("NW", "NORTH WEST") THEN "North West"
WHEN Trim(UPPER(Province)) IN ("WC", "WESTERN CAPE") THEN "Western Cape"
WHEN Trim(UPPER(Province)) IN ("MP", "MPUMALANGA") THEN "Mpumalanga"
WHEN Trim(UPPER(Province)) IN ("LP", "LIMPOPO") THEN "Limpopo"
WHEN Trim(UPPER(Province)) IN ("KZN", "KWAZULU-NATAL", "KWAZULU NATAL") THEN "KwaZulu-Natal"
END,
(SELECT mode(trim(CASE
WHEN Trim(UPPER(Province)) IN ("GP", "GAUTENG") THEN "Gauteng"
WHEN Trim(UPPER(Province)) IN ("EC", "EASTERN CAPE") THEN "Eastern Cape"
WHEN Trim(UPPER(Province)) IN ("FS", "FREE STATE") THEN "Free State"
```

SQL Data Cleaning

Province cleaning:

- Removed unnecessary spaces
- Replaced missing values
- Standardize the missing values

24 hours ago (41s) 1 SQL

```sql
TRIM(coalesce(CASE
WHEN Trim(UPPER(Province)) IN ("GP", "GAUTENG") THEN "Gauteng"
WHEN Trim(UPPER(Province)) IN ("EC", "EASTERN CAPE") THEN "Eastern Cape"
WHEN Trim(UPPER(Province)) IN ("FS", "FREE STATE") THEN "Free State"
WHEN Trim(UPPER(Province)) IN ("NW", "NORTH WEST") THEN "North West"
WHEN Trim(UPPER(Province)) IN ("WC", "WESTERN CAPE") THEN "Western Cape"
WHEN Trim(UPPER(Province)) IN ("MP", "MPUMALANGA") THEN "Mpumalanga"
WHEN Trim(UPPER(Province)) IN ("LP", "LIMPOPO") THEN "Limpopo"
WHEN Trim(UPPER(Province)) IN ("KZN", "KWAZULU-NATAL", "KWAZULU NATAL") THEN "KwaZulu-Natal"
END,
(SELECT mode(trim(CASE
WHEN Trim(UPPER(Province)) IN ("GP", "GAUTENG") THEN "Gauteng"
WHEN Trim(UPPER(Province)) IN ("EC", "EASTERN CAPE") THEN "Eastern Cape"
WHEN Trim(UPPER(Province)) IN ("FS", "FREE STATE") THEN "Free State"
WHEN Trim(UPPER(Province)) IN ("NW", "NORTH WEST") THEN "North West"
WHEN Trim(UPPER(Province)) IN ("WC", "WESTERN CAPE") THEN "Western Cape"
WHEN Trim(UPPER(Province)) IN ("MP", "MPUMALANGA") THEN "Mpumalanga"
WHEN Trim(UPPER(Province)) IN ("LP", "LIMPOPO") THEN "Limpopo"
WHEN Trim(UPPER(Province)) IN ("KZN", "KWAZULU-NATAL", "KWAZULU NATAL") THEN "KwaZulu-Natal"
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS Province,

CAST(TRIM(coalesce(CASE
  WHEN Monthly_Income LIKE '%R%' THEN CAST(REGEXP_REPLACE(Monthly_Income, '[R, ]', '') AS DOUBLE)
  WHEN Monthly_Income LIKE '%,%' THEN CAST(REGEXP_REPLACE(Monthly_Income, ',', '') AS DOUBLE)
  ELSE CAST(Monthly_Income AS DOUBLE)
END,
```

SQL Data Cleaning

Monthly_Income cleaning:

- Removed unnecessary spaces
- Replaced missing values
- Standardized the values
- Change data type to DOUBLE

Join_Date cleaning:

- Removed unnecessary spaces
- Standardized the values
- Changed data type to Date

Policy_ID cleaning:

- Removed unnecessary spaces

24 hours ago (41s) 1 SQL

```sql
CAST(TRIM(coalesce(CASE
  WHEN Monthly_Income LIKE '%R%' THEN CAST(REGEXP_REPLACE(Monthly_Income, '[R, ]', '') AS DOUBLE)
  WHEN Monthly_Income LIKE '%,%' THEN CAST(REGEXP_REPLACE(Monthly_Income, ',', '') AS DOUBLE)
  ELSE CAST(Monthly_Income AS DOUBLE)
END,
(SELECT mean(Trim(CASE
  WHEN Monthly_Income LIKE '%R%' THEN CAST(REGEXP_REPLACE(Monthly_Income, '[R, ]', '') AS DOUBLE)
  WHEN Monthly_Income LIKE '%,%' THEN CAST(REGEXP_REPLACE(Monthly_Income, ',', '') AS DOUBLE)
  ELSE CAST(Monthly_Income AS DOUBLE)
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS DOUBLE) AS Monthly_Income,

CASE
WHEN substring(trim(Join_Date),3,1) = "/" THEN to_date(trim(Join_Date), 'dd/MM/yyyy')
WHEN substring(trim(Join_Date),5,1) = "-" THEN to_date(trim(Join_Date), 'yyyy-MM-dd')
WHEN substring(trim(Join_Date),5,1) = "/" THEN to_date(trim(Join_Date), 'yyyy/MM/dd')
WHEN trim(Join_Date) LIKE '__-___-____' THEN to_date(trim(Join_Date), 'dd-MMM-yyyy')
WHEN substring(trim(Join_Date),3,1) = "-" THEN to_date(trim(Join_Date), 'MM-dd-yyyy')
END AS Join_Date,

TRIM(Policy_ID) AS Policy_ID,

TRIM(initcap(Policy_Type)) AS Policy_Type,

CAST(TRIM(coalesce(CASE
  WHEN Premium_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Premium_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Premium_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Premium_Amount, ',', '') AS DOUBLE)
```

SQL Data Cleaning

Premium_Amount cleaning:

- Removed unnecessary spaces
- Replaced missing values
- Standardized the values
- Changed data type to DOUBLE

Policy_Status cleaning:

- Removed unnecessary spaces
- Standardized the values

Claim_ID cleaning:

- Removed unnecessary spaces

Claim_Date cleaning:

- Removed unnecessary spaces
- Standardized the values
- Changed data type to Date

24 hours ago (41s) 1 SQL

```sql
CAST(TRIM(coalesce(CASE
  WHEN Premium_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Premium_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Premium_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Premium_Amount, ',', '') AS DOUBLE)
  ELSE CAST(Premium_Amount AS DOUBLE)
END,
(SELECT mean(Trim(CASE
  WHEN Premium_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Premium_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Premium_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Premium_Amount, ',', '') AS DOUBLE)
  ELSE CAST(Premium_Amount AS DOUBLE)
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS DOUBLE) AS Premium_Amount,

TRIM(CASE WHEN Trim(initcap(Policy_Type)) = "Cancelled" THEN "Canceled" END) AS Policy_Status,

TRIM(initcap(Claim_ID)) AS Claim_ID,

CASE
WHEN substring(trim(Claim_Date),3,1) = "/" THEN to_date(trim(Claim_Date), 'dd/MM/yyyy')
WHEN substring(trim(Claim_Date),5,1) = "-" THEN to_date(trim(Claim_Date), 'yyyy-MM-dd')
WHEN substring(trim(Claim_Date),5,1) = "/" THEN to_date(trim(Claim_Date), 'yyyy/MM/dd')
WHEN trim(Claim_Date) LIKE '__-___-____' THEN to_date(trim(Claim_Date), 'dd-MMM-yyyy')
WHEN substring(trim(Claim_Date),3,1) = "-" THEN to_date(trim(Claim_Date), 'MM-dd-yyyy')
END AS Claim_Date,

CAST(TRIM(coalesce(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%-%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '-', '') AS DOUBLE)
```

SQL Data Cleaning

Claim_Amount cleaning:

- Removed unnecessary spaces
- Replaced missing values
- Standardized the values
- Changed data type to DOUBLE

Claim_Status cleaning:

- Removed unnecessary spaces
- Standardized the values

Fraud_Flag cleaning:

- Removed unnecessary spaces
- Standardized the values
- Replaced missing values

24 hours ago (41s) 1 SQL

```sql
CAST(TRIM(coalesce(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%-%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '-', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE)
END,
(SELECT mean(Trim(CASE
  WHEN Claim_Amount LIKE '%R%' THEN CAST(REGEXP_REPLACE(Claim_Amount, '[R, ]', '') AS DOUBLE)
  WHEN Claim_Amount LIKE '%,%' THEN CAST(REGEXP_REPLACE(Claim_Amount, ',', '') AS DOUBLE)
  ELSE CAST(Claim_Amount AS DOUBLE)
END)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS DOUBLE) AS Claim_Amount,

TRIM(CASE WHEN Trim(initcap(Claim_Status)) = "In Review" THEN "Pending"
WHEN Trim(initcap(Claim_Status)) = "Declined" THEN "Rejected"
ELSE Trim(initcap(Claim_Status))
END) AS Claim_Status,

TRIM(coalesce(CASE WHEN TRIM(Fraud_Flag) like 0 THEN 'No'
WHEN TRIM(Fraud_Flag) like 1 THEN "Yes"
WHEN TRIM(Fraud_Flag) like "Y%%" THEN "Yes"
WHEN TRIM(Fraud_Flag) like "N%" THEN "No"
WHEN TRIM(Fraud_Flag) like "N" THEN "No"
WHEN TRIM(Fraud_Flag) like "no" THEN "No"
ELSE TRIM(Fraud_Flag)
END,
(SELECT MODE(TRIM(Fraud_Flag)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS Fraud_Flag
```

SQL Data Cleaning

The cleaned table is created and displayed with it's appropriate data types.

24 hours ago (41s) 1 SQL

```
(SELECT MODE(TRIM(Fraud_Flag)) FROM workspace.dataforge.dfa_insurance_project_raw_dataset))) AS Fraud_Flag
FROM workspace.dataforge.dfa_insurance_project_raw_dataset;

SELECT * FROM workspace.dataforge.clean_insurance_data
```

See performance (2)

Show 1 additional result

Table

	Customer_ID	Age	Gender	Province	Monthly_Income	Join_Date	Policy_ID	Policy_Type
1	CUST1000	44	Male	Gauteng	26064	2020-05-30	POL5000	Home
2	CUST1001	41	Male	Free State	3500	2025-06-16	POL5001	Auto
3	CUST1002	33	Female	Limpopo	29552.92	2020-06-14	POL5002	Health
4	CUST1003	41	Female	Western Cape	3500	2024-06-10	POL5003	Auto
5	CUST1004	42	Female	Mpumalanga	3500	2024-07-15	POL5004	Home
6	CUST1005	33	Female	Limpopo	27732.53939962478	2024-04-13	POL5005	Home
7	CUST1006	38	Male	Free State	49905.01	2022-12-19	POL5006	Motor
8	CUST1007	30	Female	KwaZulu-Natal	20975.4	2021-10-14	POL5007	Life
9	CUST1008	42	Male	North West	31654.77	2022-03-28	POL5008	Home
10	CUST1009	41	Male	Limpopo	26957.76	2020-11-02	POL5009	Funeral
11	CUST1010	47	Female	Western Cape	15268.58	2021-05-18	POL5010	Motor

SQL Data Cleaning

Invalid data handling:

- Remove data where the Customer_ID is blank
- Remove data where the Age of the Customer is under 18

24 hours ago (11s) 2

```
DELETE FROM workspace.dataforge.clean_insurance_data WHERE Customer_ID = '';
DELETE FROM workspace.dataforge.clean_insurance_data WHERE Age < 18;
SELECT * FROM workspace.dataforge.clean_insurance_data
```

See performance (3)

Show 2 additional results

Table

	Customer_ID	Age	Gender	Province	Monthly_Income	Join_Date	Policy_ID	Policy_Type
1	CUST1000	44	Male	Gauteng	26064	2020-05-30	POL5000	Home
2	CUST1001	41	Male	Free State	3500	2025-06-16	POL5001	Auto
3	CUST1002	33	Female	Limpopo	29552.92	2020-06-14	POL5002	Health
4	CUST1003	41	Female	Western Cape	3500	2024-06-10	POL5003	Auto
5	CUST1004	42	Female	Mpumalanga	3500	2024-07-15	POL5004	Home
6	CUST1005	33	Female	Limpopo	27732.53939962478	2024-04-13	POL5005	Home
7	CUST1006	38	Male	Free State	49905.01	2022-12-19	POL5006	Motor
8	CUST1007	30	Female	KwaZulu-Natal	20975.4	2021-10-14	POL5007	Life
9	CUST1008	42	Male	North West	31654.77	2022-03-28	POL5008	Home
10	CUST1009	41	Male	Limpopo	26957.76	2020-11-02	POL5009	Funeral
11	CUST1010	47	Female	Western Cape	15268.58	2021-05-18	POL5010	Motor
12	CUST1011	27	Female	North West	34630.29	2023-11-29	POL5011	Auto

Dashboard and Insights

A Power BI Dashboard was built to visualize and summarize the insights from the cleaned table dataset

Dashboard Link:
https://app.powerbi.com/view?r=eyJrIjoiYTAyNTg0MDYtNDJhZi00ZmY0LTk2N2MtMDlhNWZiMGUxMjNiIiwidCI6IjczNjVmODNhLWViY2UtNGM5ZS1iYTI2LWY5ZjM5ZmZhYTk4ZSJ9

Insights were drawn mainly from these questions:

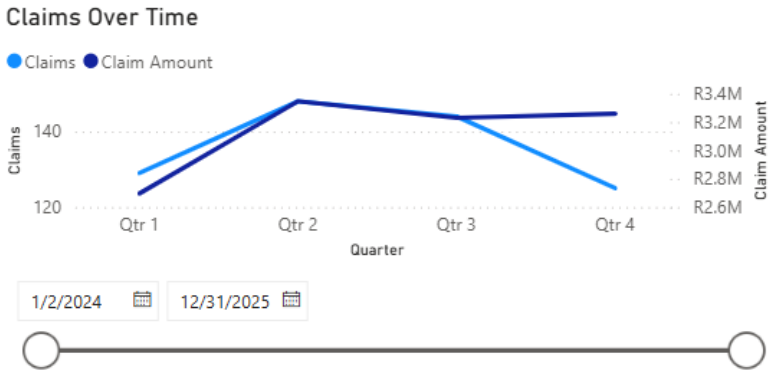
- What trends do you observe?
- Which policy type is most risky?
- Where is fraud most common?
- Which customers or segments are high-risk?

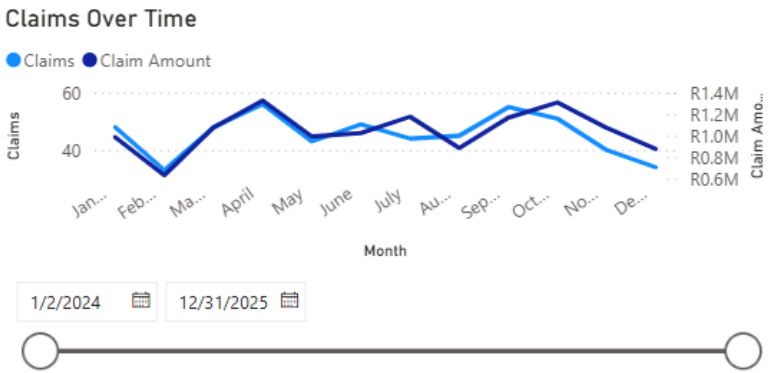
Insights

What trends do you observe?

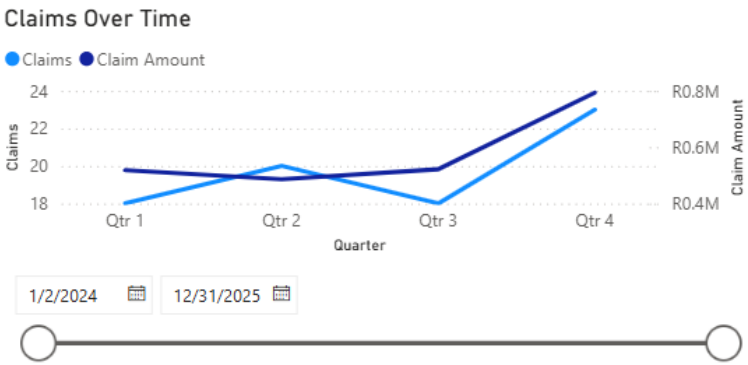
Claims have been on a downward trend in general but the fraudulent claims have been increasing on an upward trend over time along with the claim amount.

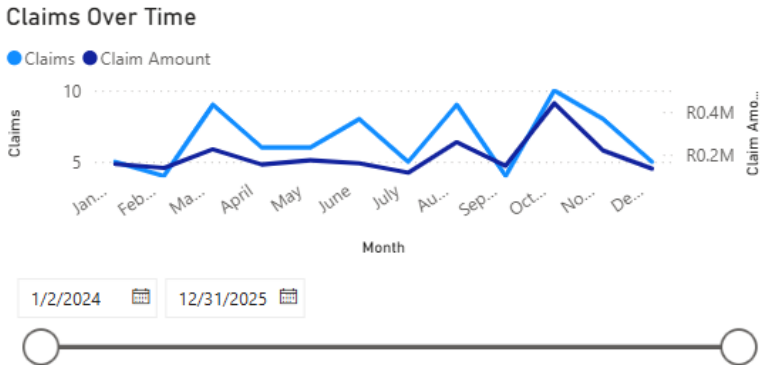
All Claims





Fraudulent Claims

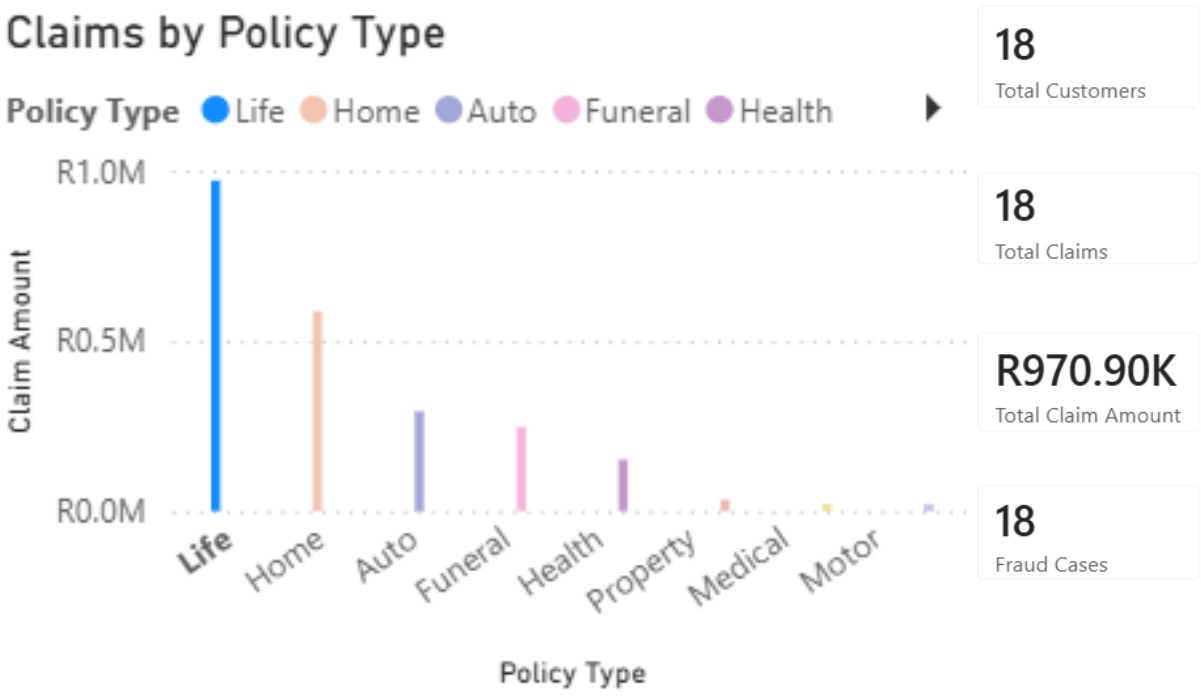




Insights

Which policy type is most risky?

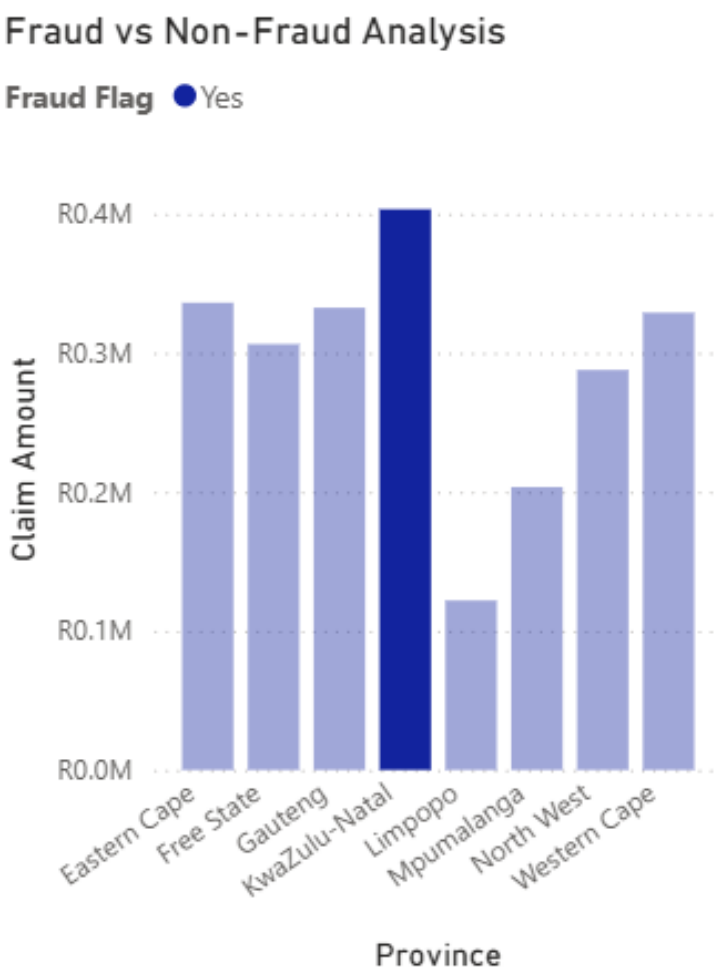
Life policy is most risky as it has 18 claims with that being the highest claim amount of R970,90k in total, 8 of the claims being approved to the amount of R531,48k.



Insights

Where is fraud most common?

Fraud is most common in KwaZulu-Natal as it has 14 fraudulent claims to the amount of R403,27k



14
Total Customers

14
Total Claims

R403.27K
Total Claim Amount

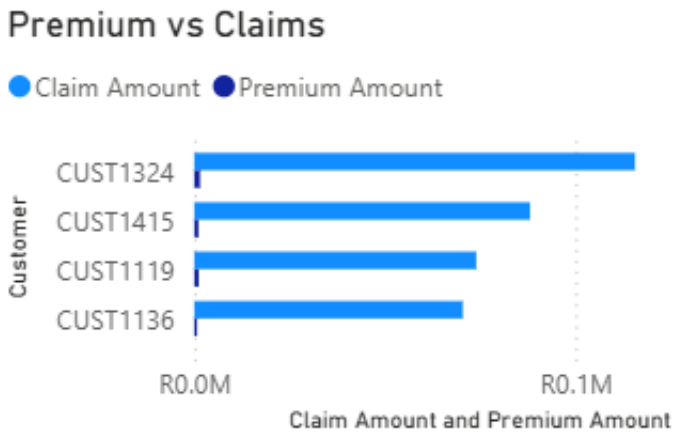
14
Fraud Cases

Insights

Which customers or segments are high-risk?

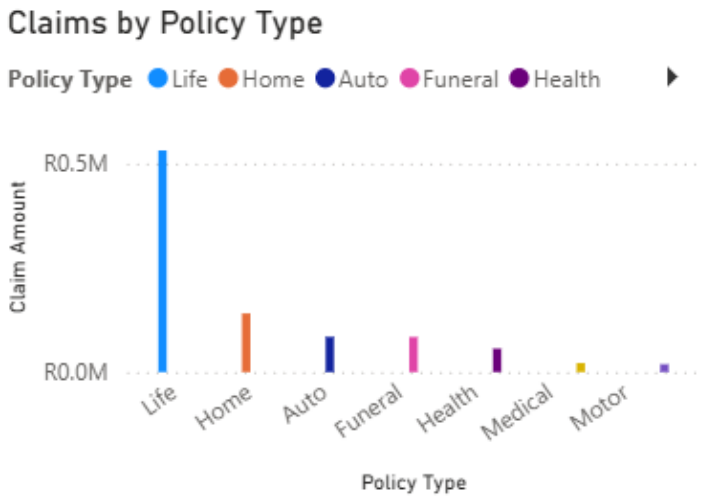
High-Risk Customers

- CUST1324 claimed 2 times to the amount of R115,58k
- CUST1415 claimed 1 time to the amount of R80,06k
- CUST1119 claimed 1 time to the amount of R73,93k
- CUST1136 claimed 1 time to the amount of R70,42k
- CUST1504 claimed 1 time to the amount of R63,46k
- CUST1220 claimed 1 time to the amount of R61,24k
- CUST1336 claimed 1 time to the amount of R58,79k



High-Risk Segments

- Life policy has 18 claims to the amount of R970,90k
- Home policy has 20 claims to the amount of R587,02k
- Auto policy has 12 claims to the amount of R292,31k
- Funeral policy has 15 claims to the amount of R246,40k
- Health policy has 11 claims to the amount of R150,80k



Recommendations

- Lower the claim amount of each high-risk segment to discourage customers from doing fraud.
- To improve customer retention, it is best to target customers within the age range of 30 years to 50 years and to target males and females earning a monthly income of around R28k.
- Make an adjustment to the premiums by increasing them to improve profitability.

THE END

THANK YOU FOR YOUR TIME.

Author

Tlale Prince Lethoko